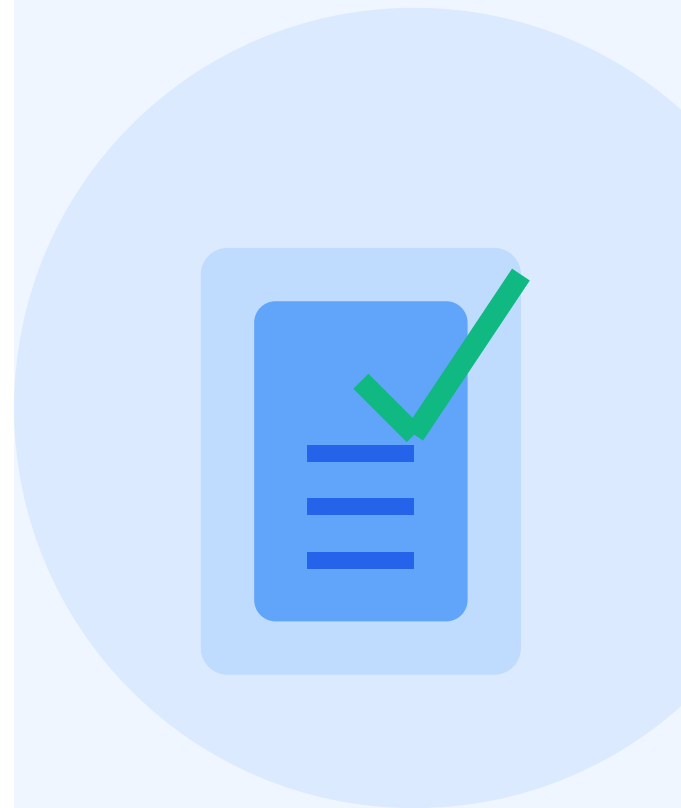


Smart Loan Risk & Approval System

Business Report

End-to-End Machine Learning Application
for Loan Risk Prediction and Approval



Report Summary

Report Type:	Business Report
Generated On:	2026-10-04 11:28
Total Records:	102
Total Columns:	21
Target Column:	Loan_Status
Active Model:	Logistic Regression

✓ Data Analysis & Insights

✓ Risk Prediction

✓ Approval Decision

✓ Better Financial Decisions

1. Dataset Overview

Total Records

102

Total Columns

21

Numeric Cols

11

Categorical Cols

10

2. Data Quality

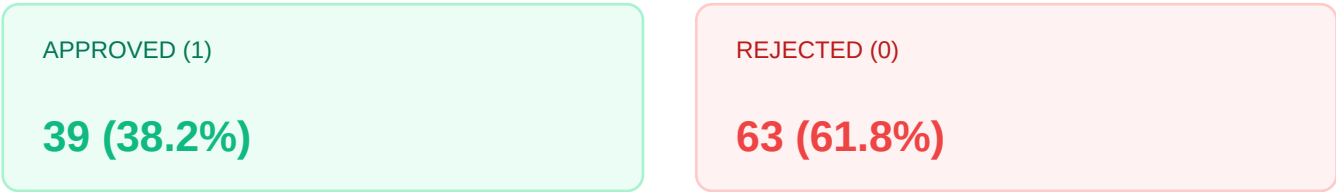
Metric	Count	Percentage
Missing Values	0	0.00%
Duplicate Rows	0	0.00%
Numerical Cols	11	52%
Categorical Cols	10	48%

✓ Dataset is ready for analysis and model training.

3. Missing Values by Column

✓ No missing values detected.
All columns have complete data.

4. Target Analysis (Loan_Status)



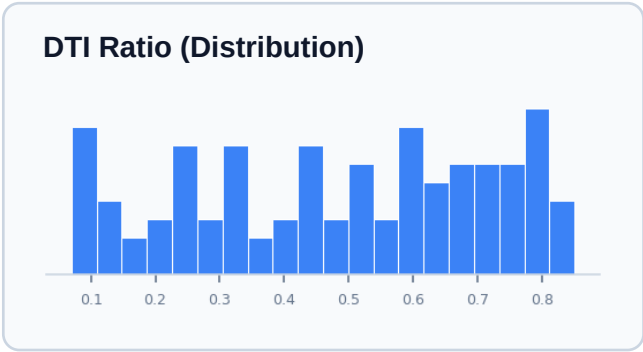
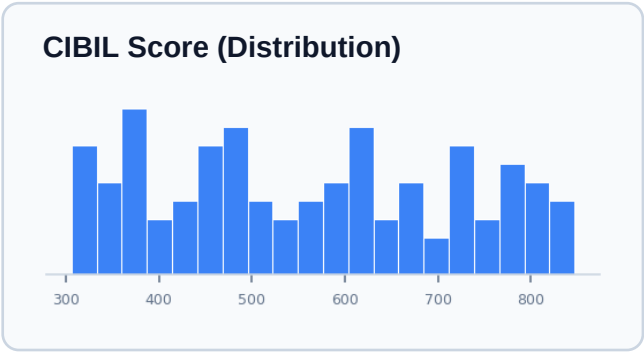
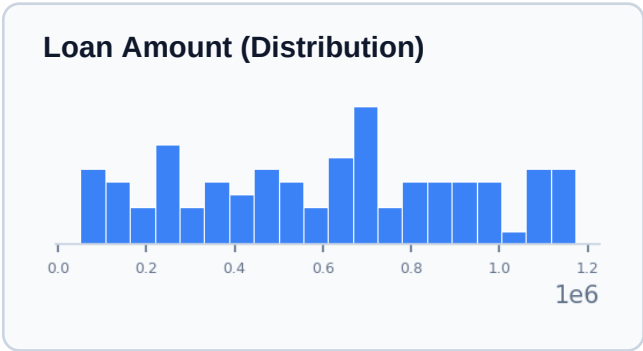
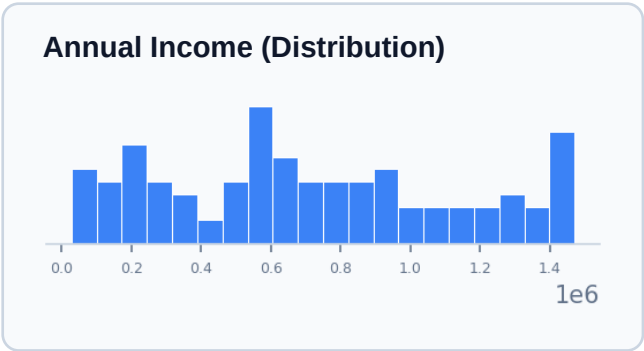
5. Descriptive Statistics

Feature	Mean	Median	Min	Max	Std Dev
Age	42.8	43.0	21.0	65.0	13.3
Employment_Years	16.1	18.0	0.0	30.0	9.1
Annual_Income	709,578.06	669,646.00	28,267.00	1,471,403.00	422,109.94
Coapplicant_Income	176,856.75	179,208.00	1,542.00	394,540.00	108,531.35
Loan_Amount	608,104.39	638,579.00	51,062.00	1,172,262.00	319,758.03
Loan_Term_Years	12.5	10.0	2.0	30.0	9.0
Cibil_Score	559.5	553.0	306.0	847.0	158.2
Existing_Loans	2.6	3.0	0.0	5.0	1.7
DTI_Ratio	0.5	0.5	0.1	0.8	0.2
Residential_Assets	2,331,195.13	2,215,685.00	11,303.00	4,988,023.00	1,515,562.25
Bank_Assets	1,034,718.12	1,080,174.00	42,534.00	1,971,164.00	551,003.19

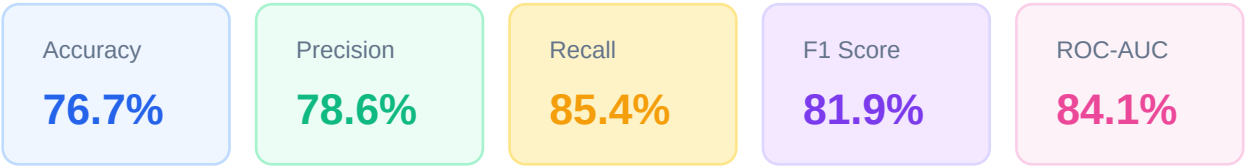
6. Outlier Analysis (IQR Method)

Feature	Outlier Count	Outlier %	Severity
No significant outliers	-	-	-

7. Distribution of Key Features

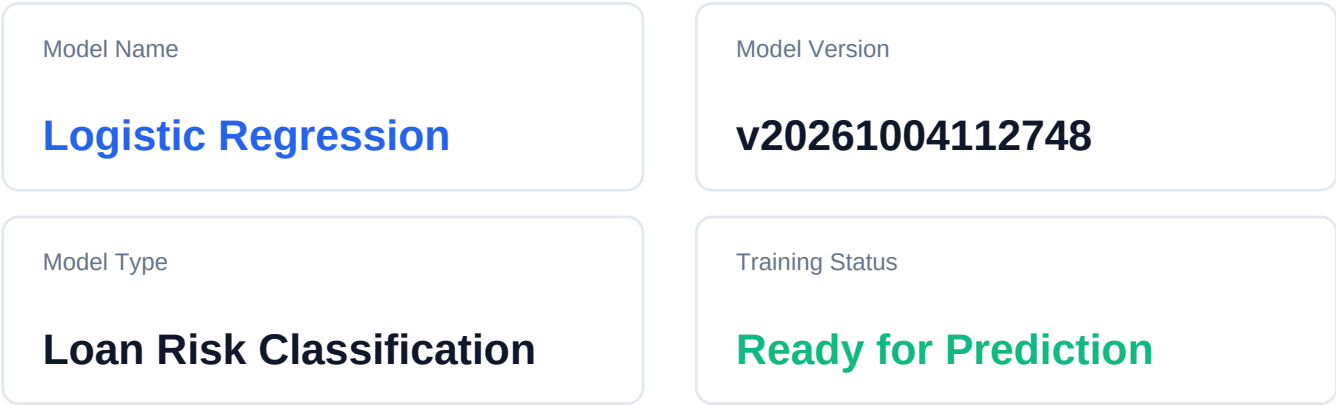


8. Model Performance



Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.767	0.786	0.854	0.819	0.841

9. Active Model Details



10. New Applicant Prediction

Applicant Information		Prediction Result	
Age	45.0	MEDIUM RISK	
Employment Years	24.0	Prediction Confidence: 52.0%	
Annual Income	855857.0	Approval Probability: 52.0%	
Coapplicant Income	234677.0	Risk Probability: 48.0%	
Loan Amount	715627.0	Decision: REVIEW	
Loan Term Years	2.0		
Cibil Score	629.0		
Existing Loans	1.0		
DTI Ratio	0.55		
Residential Assets	4176224.0		
Bank Assets	1935740.0		
Gender	Female		
Married	No		
Dependents	2		

11. Final Summary

The system analyzed 102 historical loan applications using Logistic Regression. The selected model achieved 76.7% accuracy on the evaluation dataset.

Total Records 102	Total Columns 21	Selected Model Logistic Regression
Model Accuracy 76.7%	Latest Applicant Result MEDIUM RISK	Approval Probability 52.0%